

10

NUMBER AND ALGEBRA

QUADRATIC EQUATIONS AND THE PARABOLA

Every 4 years, a few months before the Olympic Games Opening Ceremony, the Olympic flame is lit in a symbolic ceremony near the Temple of Hera, at the sacred site of the Ancient Olympics in Greece.
The flame is lit by traditional means, using the sun's rays, to demonstrate its purity.
To ensure it is lit by natural means, a parabolic mirror is used, which acts as a giant magnifying glass.
The Olympic flame has been used to light the Olympic cauldron for every Olympic Games Opening ceremony since the Summer Olympic Games held in Berlin, Germany in 1936.



Chapter outline

	Working mathematically				
10.01 Quadratic equations $ax^2 + bx + c = 0$*	U	F	PS	R	C
10.02 Completing the square*	U	F		R	C
10.03 The quadratic formula*	U	F		R	C
10.04 Which method is best?*	U	F		R	
10.05 Higher-order quadratic equations*	U	F		R	
10.06 Quadratic equation problems*	U	F	PS	R	C
10.07 Graphing quadratic equations $y = ax^2 + bx + c$*	U	F		R	C
10.08 The axis of symmetry and vertex*	U	F		R	C
10.09 Non-linear simultaneous equations*	U	F		R	C

***PATH**

Wordbank

axis of symmetry The vertical line that divides a parabola in half and passes through the parabola's vertex

discriminant The expression $b^2 - 4ac$ under the square root symbol in the quadratic formula, which determines the number of solutions in a quadratic equation

quadratic equation An equation in which the highest power of the variable is 2, for example, $2x^2 - 12x + 10 = 0$

quadratic expression An algebraic expression in which the highest power of the variable is 2; for example, $3x^2 + 4x - 6$

quadratic formula A formula for solving quadratic equations of the form $ax^2 + bx + c = 0$

solution The answer to an equation, inequality or problem, the correct value(s) of the variable that makes an equation or inequality true

vertex A turning point of a curve; a parabola has one vertex

***x*-intercept** The x value(s) at which a graph intersects the x-axis

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (7):

10.01 Quadratic equations by factorising

10.02 Solving quadratic equations
• Simple quadratic equations

10.03 The quadratic formula
• Solving quadratic equations

10.06 Quadratic equation problems

10.08 Graphing the parabola $y = ax^2+bx+c$

10.09 Non-linear simultaneous equations

Quizzes (5):

- Wordbank 10
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- Test yourself 10

Worksheets (7):

10.03 Investigating quadratic equations

10.06 Problems involving quadratic equations

10.07 Graphing parabolas 2

10.08 Features of a parabola
• A page of parabolas • Parabolas

Mind map Quadratic equations and the parabola

Puzzles (6):

10.01 Factorising quadratic equations

10.02 Complete the square order activity

10.03 Quadratic equations puzzle
• The quadratic formula

10.06 Matching parabolas with their equations

Language of maths Quadratic equations crossword

Technology (3):

10.03 The quadratic formula

10.07 Investigating parabolas 1
• Investigating parabolas 2

Spreadsheets (3):

10.03 The quadratic formula

10.07 Investigating parabolas 1
• Investigating parabolas 2

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- (PATH) solve quadratic equations of the form $ax^2 + bx + c = 0$ by factorising, completing the square and using the quadratic formula
- (PATH) investigate how the number of solutions of a quadratic equation depends on the value of the discriminant $b^2 - 4ac$
- (PATH) solve problems involving quadratic equations
- (PATH) solve higher-order quadratic equations such as $x^4 - 2x^2 - 8 = 0$
- (PATH) graph parabolas of the form $y = ax^2 + bx + c$ and identify their features
- (PATH) find the axis of symmetry and the vertex of a parabola
- (PATH) find the point of intersection of a line with a parabola, circle or hyperbola by solving linear and non-linear simultaneous equations

SkillCheck ANSWERS ON P. 657

Quiz
SkillCheck 10

1 Solve each quadratic equation.

a $4x^2 = 100$ **b** $3m^2 = 12$ **c** $x^2 - 81 = 0$

d $12k^2 = 3$ **e** $4u^2 - 49 = 0$ **f** $7w^2 - 63 = 0$

2 Factorise each expression.

a $16 - m^2$ **b** $d^2 - 121$ **c** $14y - 2y^2$

d $10p^2 + 25p$ **e** $5x^2 - 320$ **f** $18w^2 - 50$

g $k^2 + 5k + 4$ **h** $y^2 - 10y + 16$ **i** $m^2 - m - 56$

j $u^2 + 8u - 65$ **k** $w^2 - 10w + 21$ **l** $x^2 - 2x - 24$

3 If $y = x^2 + 4x - 7$, find the value of y if:

a $x = 1$ **b** $x = -1$ **c** $x = 2$ **d** $x = -\frac{1}{2}$

4 Factorise each expression.

a $3a^2 + 10a + 3$ **b** $5x^2 - 13x - 6$ **c** $6y^2 + y - 40$

d $15t^2 + 7t - 4$ **e** $5v^2 - 32v - 21$ **f** $8y^2 + 34y + 35$

g $15h^2 - 23h + 4$ **h** $12p^2 + 11p - 15$ **i** $16d^2 + 40d + 25$

Quadratic equations $ax^2 + bx + c = 0$ 10.01

PATH

Puzzle
Factorising quadratic equations

In Chapter 6, *Equations, inequalities and logarithms*, we learnt that in a **quadratic equation**, the highest power of the variable is 2. The general form of a quadratic equation is $\boldsymbol{ax^2 + bx + c = 0}$, where a, b and c are constants (numbers) and $a \neq 0$.

Solving $ax^2 + bx + c = 0$ by factorising

To solve quadratic equations of the form $ax^2 + bx + c = 0$, we need to factorise the quadratic expression on the LHS, which we learned in Chapter 4, *Products, factors and surds*).

Example 1

Solve each quadratic equation.

a $4w(3w + 2) = 0$ **b** $(5m + 2)(2m - 7) = 0$

Solution

a $4w(3w + 2) = 0$

$\therefore 4w = 0$ or $3w + 2 = 0$

$w = 0$ or $3w = -2$

$w = 0$ or $w = -\frac{2}{3}$

b $(5m + 2)(2m - 7) = 0$

$5m + 2 = 0$ or $2m - 7 = 0$

$5m = -2$ or $2m = 7$

$m = -\frac{2}{5}$ or $m = 3\frac{1}{2}$

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Video
Quadratic equations by factorising

Example 2

Solve each quadratic equation. Use substitution to check your solutions.

a $6p^2 - 9p = 0$ **b** $2x^2 - x - 15 = 0$ **c** $6y^2 = -7y + 5$

Solution

a $6p^2 - 9p = 0$

$3p(2p - 3) = 0$ Factorising

$3p = 0$ or $2p - 3 = 0$

$p = 0$ or $2p = 3$

$p = 0$ or $p = 1\frac{1}{2}$

Check: When $p = 0$, LHS $= 6 \times 0^2 - 9 \times 0 = 0 =$ RHS

When $p = 1\frac{1}{2}$, LHS $= 6 \times \left(1\frac{1}{2}\right)^2 - 9 \times 1\frac{1}{2} = 0 =$ RHS

b $2x^2 - x - 15 = 0$

$2x^2 - 6x + 5x - 15 = 0$

$2x(x - 3) + 5(x - 3) = 0$

$(x - 3)(2x + 5) = 0$

$x - 3 = 0$ or $2x + 5 = 0$

$x = 3$ or $2x = -5$

$x = 3$ or $x = -2\frac{1}{2}$

Check: When $x = 3$, LHS $= 2 \times 3^2 - 3 - 15 = 0 =$ RHS

When $x = -2\frac{1}{2}$, LHS $= 2 \times \left(-2\frac{1}{2}\right)^2 - \left(-2\frac{1}{2}\right) - 15 = 0 =$ RHS

c $6y^2 = -7y + 5$

$6y^2 + 7y - 5 = 0$

$(2y - 1)(3y + 5) = 0$

$2y - 1 = 0$ or $3y + 5 = 0$

$2y = 1$ or $3y = -5$

$y = \frac{1}{2}$ or $y = -1\frac{2}{3}$

Check: When $y = \frac{1}{2}$:

LHS $= 6 \times \left(\frac{1}{2}\right)^2 = 1\frac{1}{2}$

RHS $= -7 \times \left(\frac{1}{2}\right) + 5 = 1\frac{1}{2}$

LHS = RHS

When $y = -1\frac{2}{3}$:

LHS $= 6 \times \left(-1\frac{2}{3}\right)^2 = 16\frac{2}{3}$

RHS $= -7 \times \left(-1\frac{2}{3}\right) + 5 = 16\frac{2}{3}$

LHS = RHS

EXERCISE 10.01 ANSWERS ON P. 657

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Quadratic equations $ax^2 + bx + c = 0$ U F PS R C

EXAMPLE 1

1 Solve each quadratic equation.

a	$(m + 7)(m + 3) = 0$	**b**	$(d - 3)(d - 7) = 0$	**c**	$(y - 3)(y + 5) = 0$
d	$w(3w - 2) = 0$	**e**	$(2n + 1)(n - 3) = 0$	**f**	$(5a - 3)(2a - 1) = 0$
g	$(3x + 1)(2x + 3) = 0$	**h**	$(2c - 5)^2 = 0$	**i**	$(1 - 2f)^2 = 0$

2 What is the solution to the quadratic equation $(5x - 2)(x + 3) = 0$?
Select the correct answer **A**, **B**, **C** or **D**.

A $x = -\frac{2}{5}$ and $x = 3$ **B** $x = \frac{2}{5}$ and $x = 3$

C $x = \frac{2}{5}$ and $x = -3$ **D** $x = \frac{5}{2}$ and $x = -3$

EXAMPLE 2

3 Solve each quadratic equation.

a	$2y^2 + 7y + 6 = 0$	**b**	$2g^2 + 5g + 3 = 0$	**c**	$3d^2 + 5d + 2 = 0$
d	$5t^2 + 16t + 11 = 0$	**e**	$2p^2 - 11p + 12 = 0$	**f**	$10x^2 - 19x + 6 = 0$
g	$8y^2 - 2y - 3 = 0$	**h**	$6a^2 - 5a - 4 = 0$	**i**	$4w^2 - 7w - 15 = 0$
j	$5c^2 + 2c - 7 = 0$	**k**	$8e^2 + 10e - 3 = 0$	**l**	$3q^2 + 4q - 15 = 0$
m	$4g^2 - 20g + 25 = 0$	**n**	$18m^2 - 3m - 10 = 0$	**o**	$16 - 8w - 3w^2 = 0$

4 Express each quadratic equation in the form $ax^2 + bx + c = 0$ and solve.

a	$2x^2 = x + 15$	**b**	$4t(t + 2) = 5$	**c**	$41u = -8u^2 - 5$
d	$7m^2 = 8m - 1$	**e**	$p(p - 3) = 28$	**f**	$(e - 2)^2 = 9$
g	$t(2t - 13) = -15$	**h**	$7 = 6d^2 + 11d$	**i**	$5h^2 = 125$
j	$8f^2 = 4f$	**k**	$6w^2 + 3 = 19w$	**l**	$4a(3a + 5) - 8 = 0$

5 A certain positive number, plus its square, minus 72, equals 0.
Find the number. PS R C

10.02

10.02 Completing the square

Quadratic equations can be solved by using a method called **completing the square**. We try to make the LHS of the equation a perfect square (see Chapter 4, *Products, factors and surds*).

The method of **completing the square** is based on the following results for perfect squares.

$(x + a)^2 = x^2 + 2ax + a^2$

$(x - a)^2 = x^2 - 2ax + a^2$

We note that the last term, a^2, is the square of 'half the coefficient of x'.

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Video
Solving quadratic equations

Foundation Standard Complex

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Puzzle
Complete the square order activity

Example 3

Find the numbers that complete the square in each equation.

a $x^2 + 10x + ... = (x + ...)^2$

b $x^2 - 14x + ... = (x - ...)^2$

Solution

a The coefficient of x is 10.
Half of 10 is 5, and $5^2 = 25$.
The perfect square is
$x^2 + 10x + 25 = (x + 5)^2$

b The coefficient of x is -14.
Half of -14 is -7, and $(-7)^2 = 49$.
The perfect square is
$x^2 - 14x + 49 = (x - 7)^2$

Video
Simple quadratic equations

Example 4

Solve $(k + 3)^2 = 7$.

Solution

$(k + 3)^2 = 7$

$k + 3 = \pm\sqrt{7}$ Taking the square root of both sides

$k = \sqrt{7} - 3$ or $k = -\sqrt{7} - 3$

This solution is usually written as:

$k = -3 + \sqrt{7}$ or $k = -3 - \sqrt{7}$

Notice that the 2 solutions are surds (not rational), of the form $a + \sqrt{b}$ and $a - \sqrt{b}$.

Example 5

Solve $x^2 + 6x - 2 = 0$ by completing the square.

Solution

$x^2 + 6x = 2$ Move the constant term to the RHS

$x^2 + 6x + 3^2 = 2 + 3^2$ Halve the coefficient of x, square it and then add the square to both sides

$x^2 + 6x + 9 = 11$

$(x + 3)^2 = 11$ Express the LHS as a perfect square

$x + 3 = \pm\sqrt{11}$ Taking the square root of both sides

$x = -3 \pm \sqrt{11}$ This answer in surd form is called the **'exact answer'**

$x = -3 + \sqrt{11}$ or $x = -3 - \sqrt{11}$

Example 6

Solve $2x^2 - 3x - 4 = 0$, writing the solution correct to 2 decimal places.

Solution

$2x^2 - 3x - 4 = 0$

$2x^2 - 3x = 4$ Moving the constant term to the RHS

$x^2 - \frac{3}{2}x = 2$ Dividing both sides by 2, the coefficient of x^2

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$x^2 - \frac{3}{2}x + \left(-\frac{3}{4}\right)^2 = 2 + \left(-\frac{3}{4}\right)^2$ Completing the square

$\left(x - \frac{3}{4}\right)^2 = \frac{41}{16}$ Expressing the LHS as a perfect square and leaving the RHS as an improper fraction

$x - \frac{3}{4} = \pm\sqrt{\frac{41}{16}}$ Taking the square root of both sides

$x - \frac{3}{4} = \pm\frac{\sqrt{41}}{4}$ Simplifying the RHS

$x = \frac{3}{4} \pm \frac{\sqrt{41}}{4}$ This is the exact answer in surd form

$x = \frac{3 \pm \sqrt{41}}{4}$

$x = \frac{3+\sqrt{41}}{4}$ or $x = \frac{3-\sqrt{41}}{4}$

$x \approx 2.35$ or $x \approx -0.85$ This is an approximate answer in decimal form

EXERCISE 10.02 ANSWERS ON P. 658

Completing the square U F R C

1 Find the value of $\square$ in $x^2 - 6x + 9 = (x + \square)^2$. Select the correct answer **A**, **B**, **C** or **D**.

A -3 **B** 3 **C** -9 **D** 9

EXAMPLE 3

2 Find the numbers that 'complete the square' in each equation. R C

a $x^2 + 2x + ___ = (x + ___)^2$ **b** $p^2 - 6p + ___ = (p - ___)^2$

c $m^2 - 8m + ___ = (m - ___)^2$ **d** $k^2 + 4k + ___ = (k + ___)^2$

e $y^2 + 7y + ___ = (y + ___)^2$ **f** $w^2 - 3w + ___ = (w - ___)$

g $x^2 + x + ___ = (x + ___)^2$ **h** $h^2 - 5h + ___ = (h - ___)^2$

i $a^2 + \frac{7}{2}a + ___ = (a + ___)^2$ **j** $v^2 - \frac{5}{3}v + ___ = (v - ___)^2$

3 Find the value of $\square$ in $a^2 + \square a + \frac{49}{16} = \left(a + \frac{7}{4}\right)^2$. Select **A**, **B**, **C** or **D**.

A $\frac{7}{16}$ **B** $\frac{7}{8}$ **C** $\frac{7}{4}$ **D** $\frac{7}{2}$

EXAMPLE 4

4 Solve each equation, writing the solution in surd form. R C

a $(d + 3)^2 = 7$ **b** $(x - 5)^2 = 5$ **c** $(p + 1)^2 = 10$

d $(y - 1)^2 = 2$ **e** $\left(m - \frac{1}{2}\right)^2 = 5$ **f** $\left(t + \frac{2}{3}\right)^2 = 3$

g $(c+1)^2 = \frac{21}{2}$ **h** $(w-3)^2 = \frac{41}{2}$ **i** $\left(n + \frac{2}{3}\right)^2 = \frac{7}{9}$

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EXAMPLE 5

5 Solve each equation by completing the square. Leave your answers in exact form. R C

a	$h^2 + 2h - 5 = 0$	**b**	$r^2 - 2r - 1 = 0$	**c**	$m^2 + 6m + 2 = 0$
d	$w^2 - 4w - 60 = 0$	**e**	$a^2 - 10a - 5 = 0$	**f**	$y^2 + 8y - 3 = 0$
g	$p^2 + 12p - 5 = 0$	**h**	$x^2 - 4x + 2 = 0$	**i**	$u^2 + 9u + 14 = 0$
j	$y^2 - 3y - 8 = 0$	**k**	$b^2 - b - 5 = 0$	**l**	$q^2 - 3q + 1 = 0$
m	$2g^2 + 7g = 3$	**n**	$2x^2 = 7 - 5x$	**o**	$f(3f + 4) = 6$

EXAMPLE 6

6 Solve each equation (correct to 2 decimal places if irrational) by completing the square. R C

a	$x^2 + 12x + 9 = 0$	**b**	$m^2 - 16m - 7 = 0$	**c**	$g^2 + 4g - 3 = 0$
d	$2h^2 + 3h - 7 = 0$	**e**	$5w^2 - 4w - 3 = 0$	**f**	$3y^2 + y - 5 = 0$
g	$p(3p + 2) = 8$	**h**	$4e^2 - 4 = e$	**i**	$2n^2 = 5 - 3n$

10.03 The quadratic formula

PATH

Worksheet
Investigating quadratic equations

Puzzles
Quadratic equations puzzle

The quadratic formula

Technology
The quadratic formula

Spreadsheet
The quadratic formula

Videos
The quadratic formula

Solving quadratic equations

There is a formula for solving a quadratic equation of the form $ax^2 + bx + c = 0$.

The quadratic formula

The solutions to the quadratic equation $ax^2 + bx + c = 0$ are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example 7

Solve each quadratic equation using the quadratic formula.

a $6x^2 + x - 2 = 0$

b $3x^2 + 11x + 2 = 0$

Solution

a For $6x^2 + x - 2 = 0$:

$a = 6$, $b = 1$, and $c = -2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= \frac{-1 \pm \sqrt{1^2 - 4 \times 6 \times (-2)}}{2 \times 6}$$
$$= \frac{-1 \pm \sqrt{49}}{12}$$
$$= \frac{-1 \pm 7}{12}$$

$x = \frac{6}{12}$ or $x = \frac{-8}{12}$

$x = \frac{1}{2}$ or $x = -\frac{2}{3}$

b For $3x^2 + 11x + 2 = 0$:

$a = 3$, $b = 11$ and $c = 2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$= \frac{-11 \pm \sqrt{11^2 - 4 \times 3 \times 2}}{2 \times 3}$$
$$= \frac{-11 \pm \sqrt{97}}{6}$$

In exact surd form

Foundation Standard Complex

Example 8

Solve $2x^2 - 6 = 9x$, expressing the answer correct to 2 decimal places.

Solution

$2x^2 - 6 = 9x$

$2x^2 - 9x - 6 = 0$

Writing the quadratic equation in the form $ax^2 + bx + c = 0$

Substitute $a = 2$, $b = -9$, $c = -6$

$x = \dfrac{-(-9) \pm \sqrt{(-9)^2 - 4 \times 2 \times (-6)}}{2 \times 2}$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \dfrac{9 \pm \sqrt{129}}{4}$

$x \approx 5.09$ or $x \approx -0.59$

The discriminant

A quadratic equation can have 2, 1 or no solutions. In the quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, the expression under the square root symbol, $b^2 - 4ac$, can be used to determine the number of solutions. This expression is called the **discriminant** and has symbol Δ (the Greek capital letter 'delta').

Example 9

Use the quadratic formula to find the number of solutions of each quadratic equation.

a $2x^2 + 5x - 1 = 0$ **b** $x^2 - 16x + 64 = 0$ **c** $3x^2 + 4x + 7 = 0$

Solution

a For $2x^2 + 5x - 1 = 0$:

$a = 2$, $b = 5$ and $c = -1$

$x = \dfrac{-5 \pm \sqrt{5^2 - 4 \times 2 \times (-1)}}{2 \times 2}$

$= \dfrac{-5 \pm \sqrt{33}}{4}$

$= \dfrac{-5 + \sqrt{33}}{4}$ and $\dfrac{-5 - \sqrt{33}}{4}$

$\therefore 2x^2 + 5x - 1 = 0$ has 2 solutions.

b For $x^2 - 16x + 64 = 0$:

$a = 1$, $b = -16$ and $c = 64$

$x = \dfrac{-(-16) \pm \sqrt{(-16)^2 - 4 \times 1 \times (64)}}{2 \times 1}$

$= \dfrac{16 \pm \sqrt{0}}{2}$

$= \dfrac{16}{2}$

$= 8$

$\therefore x^2 - 16x + 64 = 0$ has 1 solution.

It's easy to see why this equation has one solution if you solve it by factorising.

c For $3x^2 + 4x + 7 = 0$:

$a = 3$, $b = 4$ and $c = 7$

$x = \dfrac{-4 \pm \sqrt{4^2 - 4 \times 3 \times 7}}{2 \times 3}$

$= \dfrac{-4 \pm \sqrt{-68}}{6}$

But $\sqrt{-68}$ is not a real number.

$\therefore 3x^2 + 4x + 7 = 0$ has no solutions.

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Notice the value of the discriminant $\Delta = b^2 - 4ac$ for each quadratic equation in **Example 9**.

Quadratic equation	$b^2 - 4ac$	Solutions	Number of solutions	Type of solutions
a $2x^2 + 5x - 1 = 0$	33	$\frac{-5 \pm \sqrt{33}}{4}$	2 because $\Delta > 0$ gives 2 answers from the quadratic formula	2 unequal real solutions
b $x^2 - 16x + 64 = 0$	0	8	1 because $\Delta = 0$ gives 1 answer from the quadratic formula	1 real solution, also called 2 equal real solutions ($x = 8, 8$)
c $3x^2 + 4x + 7 = 0$	−68	No solution	0 because $\Delta < 0$ gives 0 answers from the quadratic formula	No real solutions

'Real solutions' means solutions that are real numbers.

The discriminant $\Delta = b^2 - 4ac$

The quadratic equation $ax^2 + bx + c = 0$ has:

- 2 unequal real solutions if $b^2 - 4ac > 0$
- 1 real solution (or 2 equal real solutions) if $b^2 - 4ac = 0$
- no real solutions if $b^2 - 4ac < 0$

The discriminant 'discriminates' whether a quadratic equation has 2, 1 or 0 real solutions.

Example 10

Use the discriminant to find the number of solutions of each quadratic equation.

a $x^2 + 3x + 1 = 0$ **b** $9x^2 + 25 = 30x$

Solution

a For $x^2 + 3x + 1 = 0$:

$a = 1, b = 3$ and $c = 1$

$\Delta = b^2 - 4ac$

$= 3^2 - 4 \times 1 \times 1$

$= 5$

> 0

$\therefore x^2 + 3x + 1 = 0$ has 2 real solutions.

b For $9x^2 + 25 = 30x$:

$9x^2 - 30x + 25 = 0$

$a = 9, b = -30$ and $c = 25$

$\Delta = b^2 - 4ac$

$= (-30)^2 - 4 \times 9 \times 25$

$= 0$

$\therefore 9x^2 + 25 = 30x$ has 1 real solution (or 2 equal real solutions).

EXERCISE 10.03 ANSWERS ON P. 658

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The quadratic formula U F R C

10.03

1 Solve each quadratic equation using the quadratic formula. C

EXAMPLE 7

a $x^2 + 6x + 2 = 0$ **b** $m^2 - 5m - 3 = 0$ **c** $w^2 - 8w + 3 = 0$

d $k^2 + 3k - 5 = 0$ **e** $y^2 - 4y - 1 = 0$ **f** $p^2 + p - 5 = 0$

g $u^2 - 7u - 3 = 0$ **h** $2a^2 + 3a - 7 = 0$ **i** $5q^2 - 6q + 1 = 0$

j $3c^2 + 2c - 2 = 0$ **k** $4e^2 - 5e - 2 = 0$ **l** $3x^2 + 8x + 2 = 0$

m $2d^2 - 4d - 5 = 0$ **n** $3a^2 - 10a - 2 = 0$ **o** $2t^2 + 3t - 5 = 0$

2 Solve $2a^2 - 5a - 1 = 0$. Select the correct answer **A**, **B**, **C** or **D**.

A $a = \frac{5 \pm \sqrt{33}}{4}$ **B** $a = \frac{-5 \pm \sqrt{33}}{4}$ **C** $a = \frac{5 \pm \sqrt{17}}{4}$ **D** $a = \frac{-5 \pm \sqrt{17}}{4}$

3 Solve each quadratic equation using the quadratic formula, expressing the solution as a surd. C

a $5y^2 - 9y = 3$ **b** $3m^2 = 7 - 2m$ **c** $4x^2 = 3x + 2$

d $1 - 4k - k^2 = 0$ **e** $3m^2 - 1 - 3m = 0$ **f** $1 - 2g - 5g^2 = 0$

g $8 = 9h - 2h^2$ **h** $2w + 2 = 3w^2$ **i** $4p - 3p^2 = -1$

j $2 - 4u - 5u^2 = 0$ **k** $6a^2 = 9 - 4a$ **l** $10 = 3y + 2y^2$

4 Find the solutions to $7x^2 + 3x - 2 = 0$, correct to 2 decimal places. Select **A**, **B**, **C** or **D**.

EXAMPLE 8

A $x = 0.79, -0.36$ **B** $x = 0.36, -0.79$

C $x = 0.72, -1.58$ **D** $x = 0.82, -0.40$

5 Solve each quadratic equation, writing the solutions correct to 2 decimal places. C

a $k^2 - 9k + 1 = 0$ **b** $c^2 - 2 = 0$ **c** $m^2 - 5 = 2$

d $2n^2 + 2 = 7n$ **e** $2p^2 + 3p - 4 = 0$ **f** $6w^2 + 5w - 2 = 0$

g $3x^2 + 2 - 8x = 0$ **h** $h^2 = 7 + 2h$ **i** $1 + x - x^2 = 0$

j $t^2 = 5(t + 5)$ **k** $(x - 6)^2 = 3$ **l** $12 = 2d^2 - 3d$

6 Solve each quadratic equation and find the value of the discriminant $\Delta = b^2 - 4ac$. C

EXAMPLE 9

a $4x^2 - 5x - 2 = 0$ **b** $2k^2 + 7k - 3 = 0$ **c** $m^2 + 3m + 4 = 0$

d $3w^2 - 2w + 5 = 0$ **e** $p^2 - 18p + 81 = 0$ **f** $4x^2 - 6x + 7 = 0$

7 Use the discriminant to find the number of solutions of each quadratic equation. R C

EXAMPLE 10

a $y^2 - 9y - 2 = 0$ **b** $2m^2 + 3m - 4 = 0$ **c** $4p^2 + 5p + 13 = 0$

d $49h^2 - 70h + 25 = 0$ **e** $2n^2 - 7n + 5 = 0$ **f** $3m^2 + 7m + 4 = 0$

g $2u^2 + 4u + 2 = 0$ **h** $36 = 13a - 2a^2$ **i** $k^2 + 2k + 7 = 0$

8 If a quadratic equation has 2 real solutions, then the solutions are either rational or irrational (surds). R C

a If a quadratic equation has 2 rational solutions, what type of number is the discriminant?

b If a quadratic equation has 2 irrational solutions, what type of number is the discriminant?

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PATH

Investigation

Proving the quadratic formula

The 'completing the square' method can be used to prove the quadratic formula. Copy and complete the following proof that solves $ax^2 + bx + c = 0$ by completing the square.

$ax^2 + bx + c = 0$

$ax^2 + bx = _____$	Moving c to the RHS
$x^2 + \frac{b}{a}x = _____$	Dividing both sides by a
$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = _____ + \left(\frac{b}{2a}\right)^2$	Completing the square
$\left(x + \frac{__}{__}\right)^2 = \frac{__}{a} + \frac{__}{4a^2}$	Expressing the LHS as a perfect square
$\left(x + \frac{__}{__}\right)^2 = \frac{____}{4a^2}$	Simplifying the RHS
$\therefore x + \frac{b}{2a} = \pm\sqrt{\frac{____ + b^2}{4a^2}}$	Taking the square root of both sides
$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - ____}}{2a}$	Simplifying the RHS
$x = -\frac{__}{2a} \pm \frac{\sqrt{b^2 - ____}}{2a}$	Moving $\frac{b}{2a}$ to the RHS
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	

Technology

The quadratic formula

Set up your spreadsheet as shown below.

	A	B	C	D	E
1	a	b	c	x_1	x_2
2					
⋮					
⋮					
10					

- a, b and c are the coefficients from the quadratic equation $ax^2 + bx + c = 0$.
- x_1, x_2 are the possible solutions.

1 Enter the formula **=(-B2+SQRT(B2^2-4*A2*C2))/2*A2** in cell D2 and enter the formula **=(-B2–SQRT(B2^2-4*A2*C2))/2*A2** in cell E2.

2 Copy the formulas down to row 10.

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3 Use your spreadsheet to solve each quadratic equation.

a	$x^2 + 9x - 20 = 0$	**b**	$2x^2 + 6x + 1 = 0$	**c**	$3x^2 + 2x - 5 = 0$
d	$x^2 + 2x - 15 = 0$	**e**	$8x^2 - 10x + 3 = 0$	**f**	$x^2 + 2x - 3 = 0$
g	$4x^2 - 20x + 4 = 0$	**h**	$x^2 - 6x - 8 = 0$	**i**	$10x^2 - 3x - 1 = 0$

4 Use your spreadsheet to solve $x^2 - x + 3 = 0$. Explain your result.

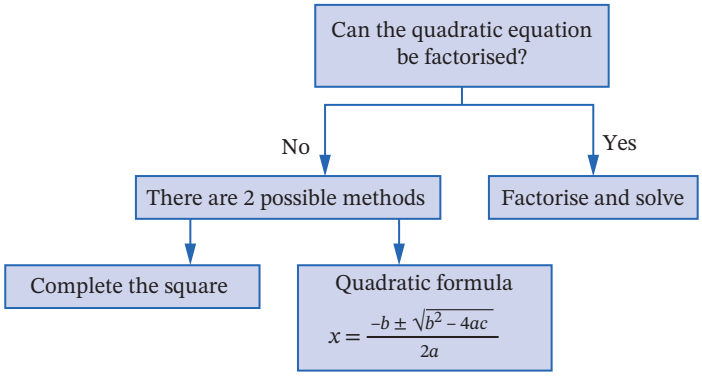
PATH

Which method is best? 10.04

PATH

We have learned 3 methods of solving quadratic equations.

Factorising is the simplest, but only some quadratic equations can be factorised, while the **quadratic formula** works for all quadratic equations.
Completing the square also works for all equations, but the algebra can be complicated.



EXERCISE 10.04 ANSWERS ON P. 658

Which method is best? U F R

1 Solve each quadratic equation using the most suitable method. You could try 2 methods to compare your answers and determine how effective the methods were. R

a	$x(2x - 3) = 0$	**b**	$m^2 = 36$	**c**	$3w^2 - 5w - 1 = 0$
d	$u^2 - u - 12 = 0$	**e**	$(5k - 3)(2k + 7) = 0$	**f**	$16 = 25a^2$
g	$x(x - 2) = 63$	**h**	$n^2 - 6n = 10$	**i**	$(a + 3)^2 = 5$
j	$(d + 3)(d - 1) = 0$	**k**	$t^2 = 4t$	**l**	$(2y - 3)^2 = 11$
m	$\frac{30}{x} - x = 7$	**n**	$2a = \frac{6 - 5a}{3a}$	**o**	$\frac{2x + 1}{2x - 1} - \frac{2x - 1}{2x + 1} = 6$

2 Solve each quadratic equation using an appropriate method, giving your answers correct to 2 decimal places where necessary. R

a	$y^2 = 5y$	**b**	$144 = a^2$	**c**	$(u - 2)^2 = 11$
d	$x^2 - 6x = -9$	**e**	$18 + 7x - x^2 = 0$	**f**	$x^2 + 10x + 5 = 0$
g	$a(2a - 1) = 7$	**h**	$2w^2 = 3w + 7$	**i**	$(5m - 1)^2 = 18$
j	$(3 - 2k)^2 = 27$	**k**	$\frac{36}{x} + 1 = 2x$	**l**	$\frac{h}{3h + 2} = \frac{h - 3}{4h}$

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10.05 Higher-order quadratic equations

PATH

Higher-order equations have the highest power of the variable being greater than 2. Equations such as $x^4 - 6x^2 + 8 = 0$, $2y^6 + 15y^3 - 8 = 0$ are examples of higher-order 'quadratic' equations. These equations can be 'simplified' or reduced to a quadratic equation.

Example 11

Solve each quadratic equation.

a $x^4 - 11x^2 + 18 = 0$ **b** $m^4 + 3m^2 - 4 = 0$ **c** $y^6 - 9y^3 - 10 = 0$

Solution

a The equation can be written as:
$(x^2)^2 - 11(x^2) + 18 = 0$.
This is a quadratic equation in x^2.
Let $u = x^2$.
$\therefore x^4 - 11x^2 + 18 = 0$ becomes
$u^2 - 11u + 18 = 0$
$(u - 9)(u - 2) = 0$
$u - 9 = 0$ or $u - 2 = 0$
$u = 9$ or $u = 2$
But $u = x^2$
$\therefore x^2 = 9$ or $x^2 = 2$
$x = \pm 3$ or $x = \pm\sqrt{2}$
The solutions of the equation are:
$x = \pm\sqrt{2}, \pm 3$

This quadratic equation has 4 solutions: $x = \sqrt{2}, -\sqrt{2}, 3$ or -3.
You could substitute back to check.

b $(m^2)^2 + 3(m^2) - 4 = 0$ is a quadratic equation in m^2.
Let $u = m^2$
$\therefore m^4 + 3m^2 - 4 = 0$ becomes
$u^2 + 3u - 4 = 0$
$(u + 4)(u - 1) = 0$
$\therefore u + 4 = 0$ or $u - 1 = 0$
$u = -4$ or $u = 1$
But $u = m^2$
$\therefore m^2 = -4$ or $m^2 = 1$
which has no solution or $m = \pm 1$
The solutions of the equation are
$m = \pm 1$.

This quadratic equation has 2 solutions: $m = 1$ or -1.
You could substitute back to check.

c $(y^3)^2 - 9(y^3) - 10 = 0$ is a quadratic equation in y^3.
Let $u = y^3$
$\therefore y^6 - 9y^3 - 10 = 0$ becomes
$u^2 - 9u - 10 = 0$
$(u + 1)(u - 10) = 0$
$\therefore u + 1 = 0$ or $u - 10 = 0$
$u = -1$ or $u = 10$
But $u = y^3$
$\therefore y^3 = -1$ or $y^3 = 10$
$y = -1$ or $y = \sqrt[3]{10}$
The solutions of the equation are
$y = -1$ and $\sqrt[3]{10}$.

This quadratic equation has 2 solutions.

EXERCISE 10.05 ANSWERS ON P. 659

PATH

Higher-order quadratic equations U F R

EXAMPLE 11

10.05

1 Use the substitution given to solve each quadratic equation. R

a	$y^4 - 12y^2 + 32 = 0$	let $u = y^2$
b	$m^4 - 13m^2 - 48 = 0$	let $u = m^2$
c	$w^4 - 8w^2 + 12 = 0$	let $u = w^2$
d	$m^6 - 9m^3 + 8 = 0$	let $u = m^3$
e	$k^6 - 26k^3 - 27 = 0$	let $u = k^3$
f	$w^6 - 4w^3 - 32 = 0$	let $u = w^3$
g	$(3x + 5)^2 - 8(3x + 5) + 12 = 0$	let $u = 3x + 5$
h	$(5y - 1)^2 - (5y - 1) - 30 = 0$	let $u = 5y - 1$
i	$(2a + 7)^2 - 10(2a + 7) + 24 = 0$	let $u = 2a + 7$
j	$p^6 + 6p^3 + 5 = 0$	let $u = p^3$
k	$8g^6 - 65g^3 + 8 = 0$	let $u = g^3$
l	$125c^6 + 999c^3 - 8 = 0$	let $u = c^3$

2 Solve each equation, correct to one decimal place where necessary. R

a	$2m^4 - 19m^2 + 24 = 0$	**b**	$5x^4 - 14x^2 + 8 = 0$
c	$7a^6 - 9a^3 - 10 = 0$	**d**	$2y^6 + y^3 - 10 = 0$
e	$3w^6 - 16w^3 - 35 = 0$	**f**	$12e^4 + 5e^2 - 2 = 0$
g	$w^4 - 21w^2 - 100 = 0$	**h**	$x^6 + 6x^3 - 16 = 0$
i	$k^6 - 64 = 0$	**j**	$m^4 - 4m^2 + 4 = 0$
k	$2y^4 + 7y^2 - 15 = 0$	**l**	$v^6 + 5v^3 - 24 = 0$

Mental skills 10: Maths without calculators ANSWERS ON P. 659

Quiz
Mental skills 10

The unitary method with percentages

1 Study this example.

If 15% of an amount is \$90, what is the whole amount?

15% of the amount = \$90

$\therefore$ 1% of the amount = \$90 ÷ 15 = \$6

$\therefore$ 100% of the amount = \$6 × 100 = \$600

The amount is \$600. *Check*: 15% × \$600 = \$90

2 Find the whole amount if:

a	5% of the amount is \$35	**b**	11% of the amount is \$88
c	20% of the amount is 80	**d**	6% of the amount is 42
e	90% of the amount is \$270	**f**	15% of the amount is \$60
g	40% of the amount is 100	**h**	120% of the amount of \$360
i	25% of the amount is \$75	**j**	8% of the amount is 40

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10.06 Quadratic equation problems

PATH

Worksheet
Problems involving quadratic equations

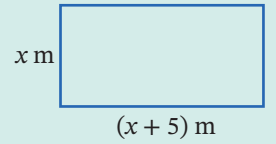
Example 12

A rectangular garden is 5 m longer than it is wide. The area of the rectangle is 84 m^2. Find the dimensions of the garden.

Solution

Let the width of the rectangular garden be x cm.

$\therefore$ The length of the garden is $(x + 5)$ cm.



Area = length × width

$= x(x + 5)$ m^2

$\therefore x(x + 5) = 84$

$x^2 + 5x = 84$

$x^2 + 5x - 84 = 0$

$(x + 12)(x - 7) = 0$

$x + 12 = 0$ or $x - 7 = 0$

$\therefore x = -12$ or $x = 7$

Since x represents a measurement of width, $x \neq -12$.

$\therefore$ The width is 7 m and the length is $7 + 5 = 12$ m

Check: Area = 7 m × 12 m = 84 m^2

Example 13

Video
Quadratic equation problems

A ball is thrown upwards and its height, h metres, after t seconds is given by the formula

$$h = 30t - 5t^2$$

At what times did the ball reach a height of 24 m? Express the answer correct to 2 decimal places.


Science Photo Library/ Alamy Stock Photo

Solution

When $h = 24$,

$24 = 30t - 5t^2$

Rearranging the equation,

$5t^2 - 30t + 24 = 0$

Using the quadratic formula,

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-30) \pm \sqrt{(-30)^2 - 4 \times 5 \times 24}}{2 \times 5}$$

$$= \frac{30 \pm \sqrt{420}}{10}$$

$t = \frac{30 + \sqrt{420}}{10}$ or $t = \frac{30 - \sqrt{420}}{10}$

$t \approx 5.05$ or $t \approx 0.95$

The ball reaches a height of 24 m at 0.95 s and 5.05 s.

Why are there 2 answers to this problem? Why is the ball at a height of 24 m at 2 different times?

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EXERCISE 10.06 ANSWERS ON P. 659

PATH

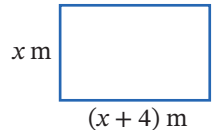
Quadratic equation problems U F PS R C

To answer each problem, first form a quadratic equation and then solve it.

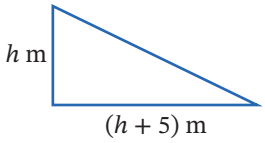
EXAMPLE 12

10.06

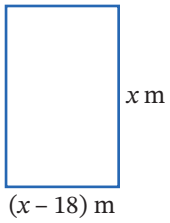
1 A garden is in the shape of a rectangle and its length is 4 m longer than its width. If the area of the garden is 96 m², find the dimensions of the garden. PS R C



2 A park is in the shape of a right-angled triangle, with its base 5 m longer than its perpendicular height. If the area of the park is 700 m², find the dimensions of the park. PS R C



3 A rectangular block of land has its width 18 m shorter than its length. If the area of the block is 1008 m², find the dimensions of the block of land. PS R C



4 A block of land is in the shape of a rectangle with its width 20 m shorter than its length. If the area of the block is 2204 m², find the length and width of the block of land. PS R C

5 A rectangular garden bed is twice as long as it is wide. Its area is 84.5 m². Find the length of the garden bed. PS R C

6 Find the dimensions of a rectangle whose length is 3 m longer than its width and its area is 180 m². PS R C

EXAMPLE 13

7 A golf ball is thrown upwards and its height, h metres, after t seconds is given by the formula $h = 18t - 5t^2$. At what times did the ball reach a height of 8 m? Answer correct to 2 decimal places. PS R C

8 After jumping from a plane, the height (in metres) of a skydiver above the ground is given by $h = 4000 - 5t^2$, where t is the time (in seconds) after jumping. R C

a How high was the plane at the moment the skydiver jumped?

b What was the skydiver's height after 20 seconds?

c The skydiver opened his parachute at 1000 m. How long did it take the skydiver to reach this height? (Give your answer correct to one decimal place.)



Maxwell.Roche/Shutterstock.com



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PATH

9 A ball is thrown from a balcony and its height (in metres) after t seconds is given by the formula $h = 30 + 12t - 5t^2$. R C

a What is the height of the ball after 2 seconds?

b When does the ball hit the ground? Answer correct to one decimal place.

c At what time (correct to one decimal place) is the ball at a height of:

i 35 m? **ii** 10 m?

10 The sum of a number and its square is 72. What is the number? PS R C

11 The product of 2 consecutive integers is 600. Find the integers. PS R C

12 When a number is subtracted from its square, the result is 1190. What is the number? PS R C

13 The difference between 2 positive integers is 12 and their product is 405. Find the integers. PS R C

14 The production costs, $\$C$, of a factory producing n toy boats each week is given by:

$$C = 0.05n^2 - 12n + 2700$$

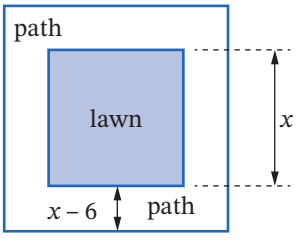
Find: R C

a the cost of producing 500 toy boats

b the profit made on 500 toy boats if each one sells for \$29.50

c the (whole) number of toy boats that can be produced at a production cost of \$8150.

15 In a garden, a square lawn area is surrounded by a path of width 6 metres less than the side of the square lawn. The area of the path is 32 square metres. Find the area of the lawn. PS R C



10.07 Graphing quadratic equations $y = ax^2 + bx + c$

PATH

Worksheet Graphing parabolas 2

Puzzle Matching parabolas with their equations

We have already graphed parabolas of the form $y = ax^2 + c$ and $y = a(x - b)^2 + c$ in Chapter 7, *Graphing curves*. Now we will graph parabolas of the general form $y = ax^2 + bx + c$.

Example 14

For each quadratic equation:

i complete a table of values and draw the graph

ii find the x-intercepts

iii find the y-intercept

iv solve $y = 0$ and compare the solutions to the x-intercepts.

a $y = x^2 + 6x + 5$ **b** $y = 2x^2 - x - 10$

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Solution

a **i** $y = x^2 + 6x + 5$

x	−6	−5	−4	−3	−2	−1	0	1
y	5	0	−3	−4	−3	0	5	12

ii The x-intercepts are -5 and -1.

iii The y-intercept is 5.

Note that this is the constant term, $c = 5$, in $y = x^2 + 6x + 5$.

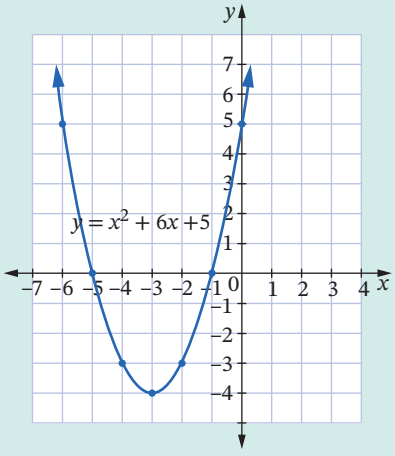
iv $x^2 + 6x + 5 = 0$

$(x + 5)(x + 1) = 0$

$x + 5 = 0$ or $x + 1 = 0$

$x = -5$ or $x = -1$

The solutions to the quadratic equation are the same as the x-intercepts of the graph of $y = x^2 + 6x + 5$.



b **i** $y = 2x^2 - x - 10$

x	−3	−2	−1	0	1	2	3
y	11	0	−7	−10	−9	−4	5

ii The x-intercepts are -2 and $2\frac{1}{2}$.

iii The y-intercept is -10.

Note that this is the constant term, $c = -10$, in $y = 2x^2 - x - 10$.

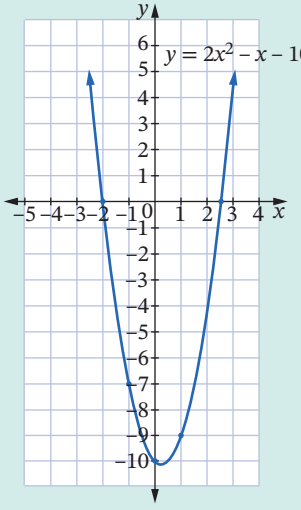
iv $2x^2 - x - 10 = 0$

$(2x - 5)(x + 2) = 0$

$2x - 5 = 0$ or $x + 2 = 0$

$x = 2\frac{1}{2}$ or $x = -2$

The solutions to the quadratic equation are the same as the x-intercepts of the graph of $y = 2x^2 - x - 10$.



The parabola $y = ax^2 + bx + c$

- If $a > 0$ (positive), the parabola is **concave up**
- If $a < 0$ (negative), the parabola is **concave down**
- The ***y*-intercept** of the parabola is c
- The ***x*-intercepts** of the parabola are the solutions to the quadratic equation $ax^2 + bx + c = 0$

PATH

Technology
Investigating parabolas 1
Investigating parabolas 2

Spreadsheets
Investigating parabolas 1
Investigating parabolas 2

10.07

PATH

Example 15

Graph each quadratic equation, showing its x- and y-intercepts.

a $y = x(2x - 5)$ **b** $y = 2x^2 + x - 6$ **c** $y = -2x^2 - 3x + 9$

Solution

a $y = x(2x - 5)$

$= 2x^2 - 5x$

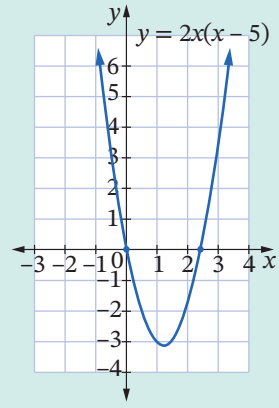
$a = 2 > 0$, so the parabola is concave up

x-intercepts: $x(2x - 5) = 0$

$x = 0$ and $2x - 5 = 0$

$x = 0$ and $x = 2\frac{1}{2}$

y-intercept: $c = 0$



b $y = 2x^2 + x - 6$

$a = 2 > 0$, so the parabola is concave up

x-intercepts:

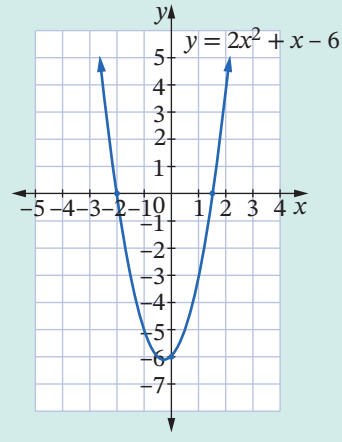
$2x^2 + x - 6 = 0$

$(2x - 3)(x + 2) = 0$

$2x - 3 = 0$ and $x + 2 = 0$

$x = 1\frac{1}{2}$ and $x = -2$

y-intercept: $c = -6$



c $y = -2x^2 - 3x + 9$

$a = -2 < 0$, so the parabola is concave down

x-intercepts:

$-2x^2 - 3x + 9 = 0$

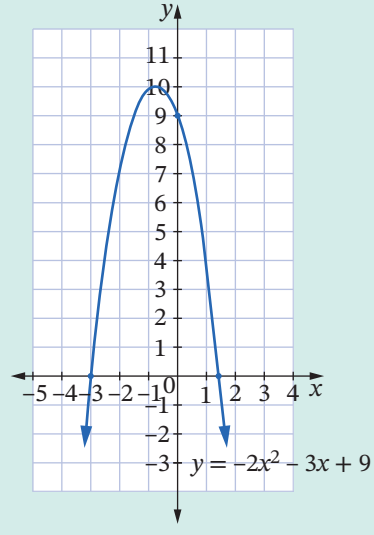
$2x^2 + 3x - 9 = 0$ Dividing by –1

$(2x - 3)(x + 3) = 0$

$2x - 3 = 0$ and $x + 3 = 0$

$x = 1\frac{1}{2}$ and $x = -3$

y-intercept: $c = 9$



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EXERCISE 10.07 ANSWERS ON P. 659

PATH

Graphing quadratic equations $y = ax^2 + bx + c$ U F R C

EXAMPLE 14

10.07

1 For each quadratic equation: R C

i complete a table of values and draw the graph

ii find the x-intercepts

iii find the y-intercept

iv solve $y = 0$ and compare the solutions to the x-intercepts

a $y = x^2 + 4x + 3$ **b** $y = 2x^2 + 7x + 3$

c $y = 2x^2 - 3x - 9$ **d** $y = x^2 - 2x$

2 Find the y-intercept of the parabolas with equation:

a $y = 3x^2 - 2x - 5$ **b** $y = 2x^2 + 6x + 3$ **c** $y = 5x^2 - 10x$

3 Which quadratic equation below represents a parabola with x-intercepts 2 and –5? Select the correct answer **A**, **B**, **C** or **D**.

A $y = x^2 + 7x - 10$ **B** $y = x^2 - 7x - 10$

C $y = x^2 + 3x - 10$ **D** $y = x^2 - 3x - 10$

4 Which quadratic equation below represents a parabola with only one x-intercept, 1, and a y-intercept of 2? Select **A**, **B**, **C** or **D**.

A $y = (x - 1)(x - 2)$ **B** $y = 2(x - 1)^2$

C $y = (x + 1)(x + 2)$ **D** $y = (2x - 1)^2$

EXAMPLE 15

5 Graph each quadratic equation, showing its x- and y-intercepts.

a $y = x(x - 4)$ **b** $y = 3x(x + 2)$ **c** $y = (x + 3)(x - 5)$

d $y = (x - 2)(2x - 5)$ **e** $y = -(x + 1)(3x + 2)$ **f** $y = x^2 + 6x + 8$

g $y = -x^2 + 4x + 5$ **h** $y = 3x^2 + 11x - 20$

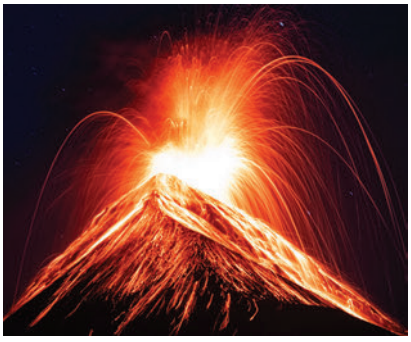
Did you know?

Parabolas in action

When an object is thrown or a bullet is fired from a gun, its path is a parabola. The length of the path will depend on the initial speed. The photo shows parabolas of lava being expelled from a volcano.

Parabola-shaped mirrors or reflectors are used in car headlights, spotlights and satellite dishes.

Investigate the property of parabolas that makes them useful as reflectors of light and radio signals.



Daniel Lopez Toriello/Shutterstock.com

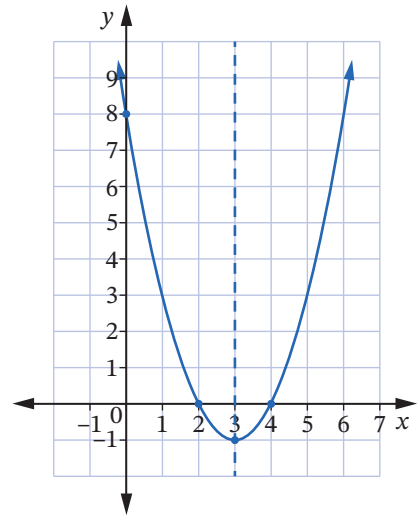
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Investigation

The axis of symmetry and vertex of a parabola

1 The graph of $y = x^2 - 6x + 8$ is shown.



- **a** Write the equation of the axis of symmetry of the parabola.
- **b** What are the x-intercepts of the parabola?
- **c** What is the midpoint of the interval joining the x-intercepts?
- **d** How can the x-intercepts be used to find the equation of the axis of symmetry?
- **e** Copy and complete: The equation of the axis of symmetry can be found by finding the _____ of the x-intercepts.

2 According to the quadratic formula, the solutions to $ax^2 + bx + c = 0$ are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. This means that the x-intercepts of the parabola $y = ax^2 + bx + c$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, that is, $x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$.
Hence find the equation of the axis of symmetry of $y = ax^2 + bx + c$.

3 **a** For $y = x^2 - 6x + 8$, what are the values of a and b?

b Find the value of $-\frac{b}{2a}$.

c How is the value of $-\frac{b}{2a}$ related to the axis of symmetry of $y = x^2 - 6x + 8$?

Technology

The vertex of the parabola

In this activity, we will use dynamic geometry software to graph parabolas and identify the vertex.

1 Graph $y = (x + 2)^2 + 6$ and find the coordinates of its vertex.

2 Graph each equation and identify its vertex.

- **a** $y = (x + 3)^2 + 7$
- **b** $y = (x + 4)^2 - 5$
- **c** $y = (x - 1)^2 + 6$
- **d** $y = (x - 5)^2 - 6$
- **e** $y = -(x + 2)^2 - 3$
- **f** $y = -(x - 6)^2 - 8$

3 Do you see a pattern? By comparing the graphs drawn and the equations identify the features that represent the vertex for each equation in question **2** above.

The axis of symmetry and vertex 10.08

The axis of symmetry and vertex of $y = ax^2 + bx + c$

- The **axis of symmetry** of the parabola is the vertical line: $x = -\frac{b}{2a}$
- The **axis of symmetry** passes through the point halfway between the 2 x-intercepts of the parabola
- The **vertex** of the parabola lies on the axis of symmetry, so its ***x*-coordinate** is $x = -\frac{b}{2a}$ and its ***y*-coordinate** can be found by substituting the x-coordinate into the equation of the parabola

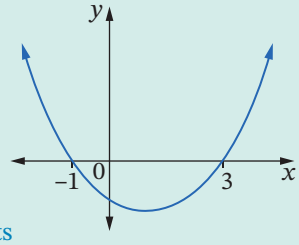
PATH

Worksheets
Features of a parabola
A page of parabolas
Parabolas

10.08

Example 16

Find the equation of the axis of symmetry of this parabola.



Solution

The x-intercepts are -1 and 3.

The equation of the axis of symmetry is:

$x = \frac{-1+3}{2}$ Finding the average of the x-intercepts

$= 1$

Example 17

A parabola has the equation $y = 2x^2 - 4x + 3$.

a Find the equation of its axis of symmetry.

b Find the coordinates of the vertex of the parabola.

Solution

a For $y = 2x^2 - 4x + 3$, $a = 2$, $b = -4$ and $c = 3$.

The axis of symmetry is:

$x = -\frac{b}{2a}$

$= -\frac{-4}{2 \times 2}$

$= 1$

$x = 1$ is the equation of the axis of symmetry.

b The vertex lies on the axis of symmetry.

Substitute $x = 1$ in $y = 2x^2 - 4x + 3$

$y = 2 \times 1^2 - 4 \times 1 + 3$

$= 1$

The vertex is (1, 1).

PATH

Video
Graphing the parabola $y = ax^2 + bx + c$

Example 18

Sketch the parabola $y = 2x^2 + 5x + 1$.

Solution

$a = 2 > 0$, so the parabola is concave up.

y-intercept $= 1$.

$y = 2x^2 + 5x + 1$ cannot be factorised, so the x-intercepts are not rational.

For the vertex:

$x = -\dfrac{b}{2a}$ where $a = 2$, and $b = 5$

$= -\dfrac{5}{2 \times 2}$

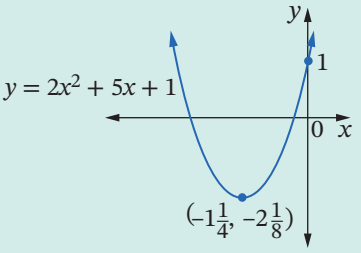
$= -1\dfrac{1}{4}$

Substitute $x = -1\dfrac{1}{4}$ into $y = 2x^2 + 5x + 1$

$y = 2 \times \left(-1\dfrac{1}{4}\right)^2 + 5 \times \left(-1\dfrac{1}{4}\right) + 1$

$= -2\dfrac{1}{8}$

$\therefore$ The vertex is $\left(-1\dfrac{1}{4}, -2\dfrac{1}{8}\right)$.



Graphing the parabola $y = ax^2 + bx + c$

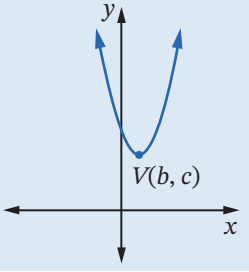
- Use the sign of a to determine whether the parabola is **concave up** or **down**
- Find the ***y*-intercept**, c
- Find the ***x*-intercepts** where possible by solving a quadratic equation
- Use $x = -\dfrac{b}{2a}$ to find the **axis of symmetry** and the **vertex**

The vertex equation of a parabola $y = (x - b)^2 + c$

In Chapter 7, *Graphing curves*, we learned about the graph of $y = a(x - b)^2 + c$.

The parabola $y = a(x - b)^2 + c$

- The parabola $y = a(x - b)^2 + c$ is the graph of $y = ax^2$ translated b units to the right (or left if b is negative) and c units up (or down if c is negative)
- The parabola $y = a(x - b)^2 + c$ has its vertex at (b, c).
  $y = a(x - b)^2 + c$ is called the **vertex form** of the quadratic equation.



Just as a circle with equation $(x - a)^2 + (y - b)^2 = r^2$ has its centre at (a, b), a parabola with equation $y = a(x - b)^2 + c$ has its vertex at (b, c).

Example 19

Find the vertex of the parabola with equation:

a $y = (x - 1)^2 + 3$ **b** $y = -2(x + 3)^2 - 5$

c $y = x^2 + 8x + 5$ by completing the square

Solution

a $y = (x - 1)^2 + 3$

Comparing against $y = (x - b)^2 + c$, this means $b = 1$ and $c = 3$, so the vertex is at $(1, 3)$.

b $y = -2(x + 3)^2 - 5$

$b = -3$ and $c = -5$ so the vertex is at $(-3, -5)$.

c Complete the square to get the equation in the form $y = (x - b)^2 + c$.

$y = x^2 + 8x + 5$

$= (x^2 + 8x + 4^2) - 4^2 + 5$ Completing the square on $x^2 + 8x$

$= (x^2 + 8x + 16) - 16 + 5$

$= (x + 4)^2 - 11$

$b = -4$ and $c = -11$, so the vertex is at $(-4, -11)$

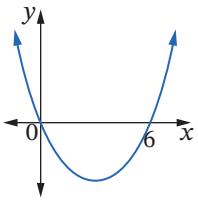
EXERCISE 10.08 ANSWERS ON P. 660

The axis of symmetry and vertex U F R C

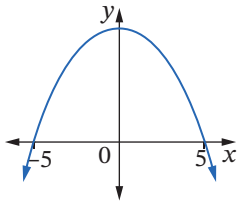
1 Write the equation for the axis of symmetry in each parabola. R C

EXAMPLE 16

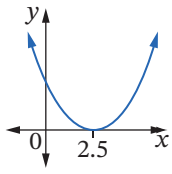
a



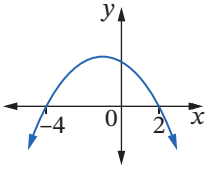
b



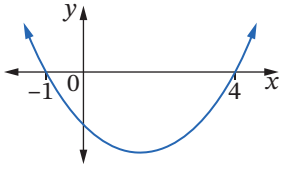
c



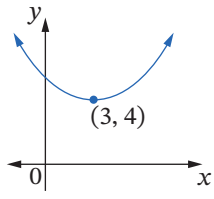
d



e



f



2 For each parabola with the equation given below:

i find the equation of its axis of symmetry

ii find the coordinates of its vertex

EXAMPLE 17

a $y = x^2 - 6x + 8$ **b** $y = -x^2 + 10x - 9$ **c** $y = x^2 - 2x + 10$

d $y = -x^2 + 8x + 9$ **e** $y = -x^2 + x - 25$ **f** $y = 5x^2 + 40x$

g $y = 24x - 3x^2$ **h** $y = 4x^2 + 2x - 1$ **i** $y = 1 - 3x - 9x^2$

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EXAMPLE 18

3 Graph each quadratic equation, showing:

- **i** the x-intercepts (correct to one decimal place where necessary)
- **ii** the y-intercept
- **iii** the equation of the axis of symmetry
- **iv** the coordinates of the vertex
- **v** the concavity

a $y = x^2 - 6x - 40$ **b** $y = x^2 - 3x$ **c** $y = 2x^2 + 3x + 4$

d $y = -x^2 + 6x + 5$ **e** $y = -4x^2 - 12x + 21$ **f** $y = x^2 - 8x + 3$

g $y = 5x^2 + 7x + 4$ **h** $y = 8x - 2x^2$ **i** $y = -2x^2 + 7x - 6$

4 Calculate the value of the discriminant $\Delta = b^2 - 4ac$ of each quadratic equation in **Question 3** and use it to state the number of x-intercepts its graph has. R C

5 State whether the discriminant is positive, negative or 0 for the equation of each parabola in **Question 1**. R C

EXAMPLE 19

6 Which one of these equations is of a parabola with vertex (–2, 9)?
Select the correct answer **A**, **B**, **C** or **D**.

A $y = (x + 2)^2 + 9$ **B** $y = (x - 2)^2 + 9$

C $y = (x + 2)^2 - 9$ **D** $y = (x - 2)^2 - 9$

7 Find the vertex of the parabola with equation:

a $y = (x + 2)^2 + 3$ **b** $y = (x + 5)^2 - 2$ **c** $y = (x - 3)^2 + 1$

d $y = -(x - 1)^2 + 5$ **e** $y = -(x + 7)^2 - 2$ **f** $y = -(x - 10)^2 - 8$

8 By completing the square, find the vertex of the parabola with each equation.
Check your answers using the $x = -\frac{b}{2a}$ method.

a $y = x^2 + 4x + 1$ **b** $y = x^2 + 5x - 4$ **c** $y = -x^2 + 2x + 5$

d $y = x^2 - 6x$ **e** $y = 2 - 2x + x^2$ **f** $y = 2x^2 - 5x + 2$

Technology

Graphing parabolas

1 Use dynamic geometry software to graph $y = x^2 + 5x + 6$.

a What is the axis of symmetry?

b What is the vertex?

2 Graph $y = 2x^2 - 9x - 35$ and find its axis of symmetry and vertex.

3 Enter each equation and identify its axis of symmetry and vertex.

a $y = x^2 - 14x + 49$ **b** $y = 2x^2 - 11x - 21$ **c** $y = 8x^2 + 28x + 20$

d $y = 8 + 2x - x^2$ **e** $y = 20 - x - x^2$ **f** $y = 40 + 2x - 3x^2$

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Non-linear simultaneous equations 10.09

The points of intersection of a line with a parabola, circle or hyperbola may be found either graphically or algebraically. The algebraic method involves solving simultaneous equations using the substitution method, which you learned about in Chapter 9, *Simultaneous equations*.

PATH

10.09

Video
Non-linear simultaneous equations

Example 20

Find the points of intersection of:

a the line $y = 2x - 3$ and the parabola $y = x^2 - 3x + 1$

b the line $y = x + 2$ and the circle $x^2 + y^2 = 4$

c the line $y = x + 5$ and the hyperbola $y = \frac{6}{x}$

Solution

a $y = 2x - 3$ [1]

$y = x^2 - 3x + 1$ [2]

Use [2] to substitute for y in [1].

$x^2 - 3x + 1 = 2x - 3$

$x^2 - 5x + 4 = 0$

$(x - 4)(x - 1) = 0$

$x - 4 = 0$ or $x - 1 = 0$

$x = 4$ or $x = 1$

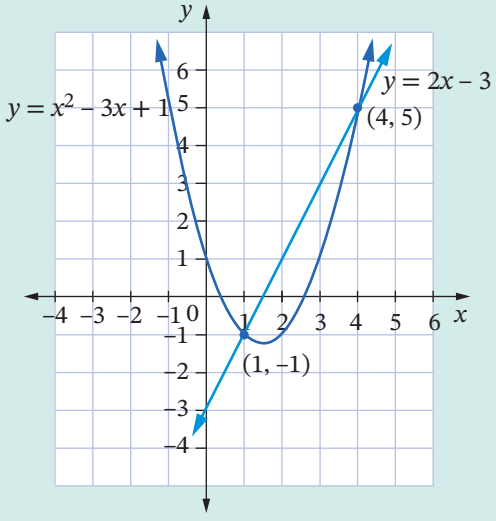
Substitute $x = 4$ and $x = 1$ into [1] to find y.

$y = 2 \times 4 - 3$ and $y = 2 \times 1 - 3$

$= 5$ $= -1$

The solutions are $x = 4, y = 5$ and $x = 1, y = -1$.

The points of intersection are (4, 5) and (1, −1).



This solution can also be found by graphing both equations and finding their points of intersection, as shown in the graph.

b $y = x + 2$ [1]

$x^2 + y^2 = 4$ [2]

Use [1] to substitute for y in [2].

$x^2 + (x + 2)^2 = 4$

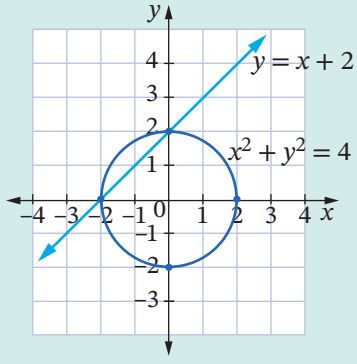
$x^2 + x^2 + 4x + 4 = 4$

$2x^2 + 4x = 0$

$2x(x + 2) = 0$

$2x = 0$ or $x + 2 = 0$

$x = 0$ or $x = -2$



PATH

Substitute $x = 0$ and $x = -2$ into [1] to find y.

$y = 0 + 2$ and $y = -2 + 2$

$y = 2$ and $y = 0$

The points of intersection are $(0, 2)$ and $(-2, 0)$.

This solution can also be found by graphing both equations and finding their points of intersection, as shown in the graph.

c $y = x + 5$ [1]

$y = \frac{6}{x}$ [2]

Use [1] to substitute for y in [2].

$$x + 5 = \frac{6}{x}$$
$$x(x + 5) = x \times \frac{6}{x}$$
$$x^2 + 5x = 6$$
$$x^2 + 5x - 6 = 0$$
$$(x + 6)(x - 1) = 0$$

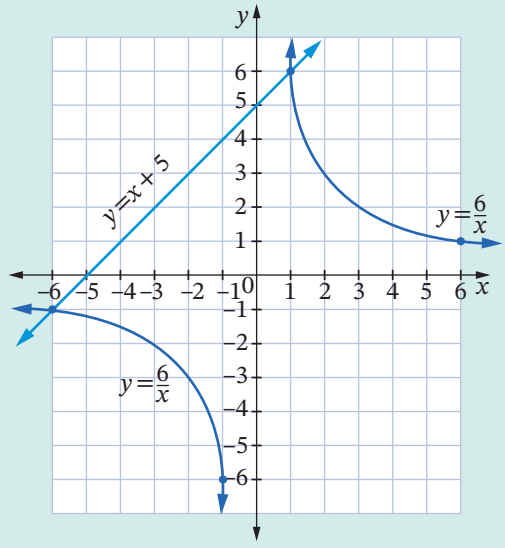
$x = -6$ or $x = 1$

Substitute $x = -6$ and $x = 1$ into [1] to find y.

$y = -6 + 5$ and $y = 1 + 5$

$y = -1$ and $y = 6$

The points of intersection are $(-6, -1)$ and $(1, 6)$. (Also, see their graphs).



EXERCISE 10.09 ANSWERS ON P. 661

Non-linear simultaneous equations UFRC

EXAMPLE 20

1 Find the points of intersection of the line and the parabola with equations: R

a $y = x^2$ and $y = 5x - 6$

b $y = 2x^2$ and $y = 8x$

c $y = x^2 + 10$ and $y = 2x + 18$

d $y = 5x^2$ and $y = x + 6$

e $y = 3x^2 + 2x + 10$ and $y = 12 - 3x$

f $y = 9x - x^2$ and $y = 6x - 10$

g $y = 6x - 3$ and $y = x^2 - 8x + 46$

h $x + y = 2$ and $y = x^2 - 2x$

2 Find the points of intersection of the line: R

a $y = x - 3$ and the hyperbola $y = -\frac{2}{x}$

b $y = 1 - x$ and the circle $x^2 + y^2 = 1$

c $x + y = 3$ and the circle $x^2 + y^2 = 9$

d $y = x + 5$ and the circle $x^2 + y^2 = 25$

e $y = 5x + 2$ and the hyperbola $y = \frac{7}{x}$

f $2x - y = 1$ and the hyperbola $y = \frac{15}{x}$

3 Explain why there are no points of intersection between the equations $y = -3x - 1$ and $y = \frac{2}{x}$. Show working to help justify your answer. R C

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Technology

Points of intersection

In this activity, we will use dynamic geometry software to find the point of intersection of a line and a curve.

1 Enter the following equations of a line and a parabola:

$$y = 3x - 4 \text{ and } y = 2x^2 + x - 16$$

2 Click on each point of intersection of the line and the curve to display its coordinates.

3 Check your answers to Exercise 10.09 by graphing the pairs of equations to find the points of intersection.

Power plus ANSWERS ON P. 661

1 **a** Show that $(3x^2 - 7x)^2 + 5(3x^2 - 7x) + 4 = 9x^4 - 42x^3 + 64x^2 - 35x + 4$.

b Hence, solve the equation $9x^4 - 42x^3 + 64x^2 - 35x + 4 = 0$.

2 Solve each equation.

a $(2^x - 32)(3^x - 27) = 0$

b $(2^x - 1)\left(2^x - \frac{1}{2}\right) = 0$

c $\left(2^x - \frac{1}{8}\right)\left(10^x - \frac{1}{100}\right) = 0$

d $2^{2x} - 2 \times 2^x + 1 = 0$

e $3^{2x} + 4 \times 3^x - 5 = 0$

f $4^x - 2^{x+2} - 32 = 0$

g $9^x - 10 \times 3^x + 9 = 0$

3 The solutions to the quadratic equation $(x - \alpha)(x - \beta) = 0$ are α and β.

a Prove that $(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta$.

b Hence find a quadratic equation of the form $ax^2 + bx + c = 0$ with solutions:

i 2 and 7

ii 5 and –8

iii –4 and $\frac{1}{2}$

iv $\sqrt{3}$ and $-\sqrt{3}$

v $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$

4 Does the line $y = 8x + 5$ intersect once, twice or completely miss the parabola $y = 2x^2 + 7x - 6$?

5 A gift box is to be made from a rectangular piece of cardboard that is 3 times as long as it is wide. 5 cm squares are cut from each corner and the ends are then folded up to make the box. If the volume of the box is 4340 cm^3, find the length and width of the piece of cardboard.

CHAPTER 10 REVIEW

Quiz
Language of maths 10

Puzzle
Quadratic equations crossword

Language of maths

axis	axis of symmetry	circle	coefficient
complete the square	concave down	concave up	discriminant
exact	factorise	hyperbola	left-hand side (LHS)
parabola	quadratic equation	quadratic formula	right-hand side (RHS)
simultaneous equations	solution	square root	substitution
surd	vertex	x-intercept	y-intercept

1 What is a **quadratic equation**?

2 What is the **coefficient** of x in $ax^2 + bx + c = 0$?

3 What are the 3 methods for solving a quadratic equation?

4 How do you write the solution to a quadratic equation in **exact form**?

5 How many solutions can a quadratic equation have?

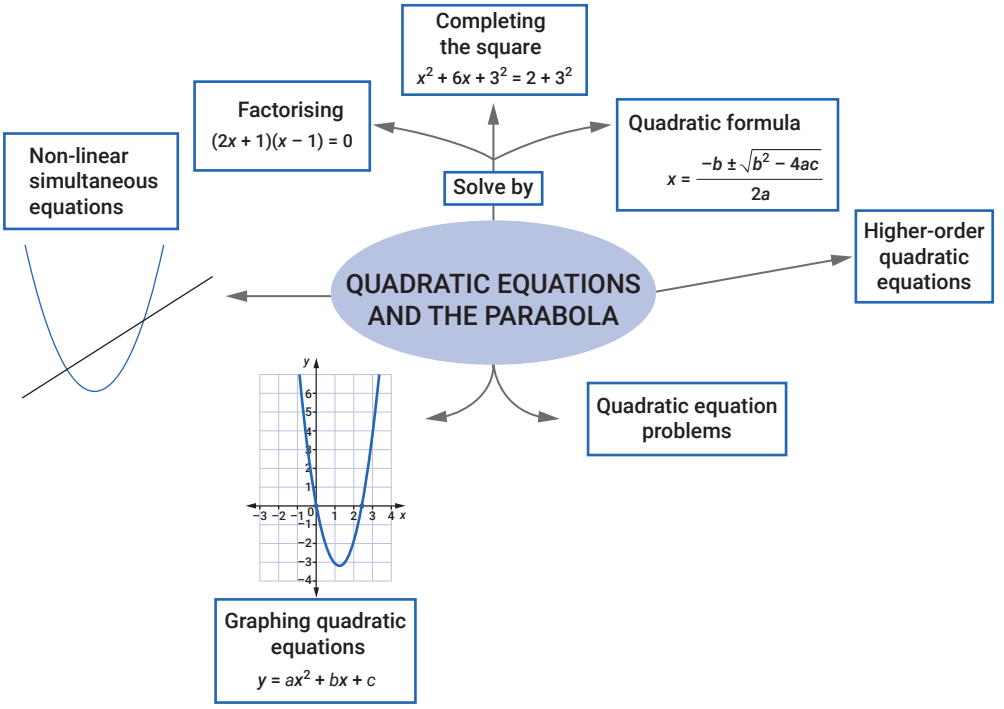
6 What are higher-order quadratic equations?

Topic summary

- Write in your own words what you have learned in this chapter.
- Write which parts of this chapter were new to you.
- What parts of this chapter did you find difficult?

Worksheet
Mind map: Quadratic equations and the parabola

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



9780170479660

TEST YOURSELF 10

ANSWERS ON P. 661

Quiz
Test yourself 10

PATH

1 Solve each quadratic equation. 10.01

a $(m-2)(2m+3)=0$ **b** $3x(x-5)=0$ **c** $p^2-2p-48=0$

d $y^2-8y+12=0$ **e** $3w^2+19w+20=0$ **f** $8k^2=2k+3$

g $n(2n-4)=30$ **h** $(2d+3)^2=36$ **i** $4x^2-15=4x$

2 Solve each quadratic equation by completing the square. 10.02

a $y^2+4y-7=0$ **b** $2p^2-3p-3=0$ **c** $5w^2+2w-1=0$

3 Solve each quadratic equation using the quadratic formula. 10.03

a $m^2+5m-3=0$ **b** $2d^2-7d-11=0$ **c** $3k^2-k=21$

4 Use the discriminant to find the number of solutions of each quadratic equation. 10.03

a $x^2+3x-1=0$ **b** $4h^2-3h+5=0$ **c** $w^2=8w-16$

5 Use the most suitable method to solve each quadratic equation. 10.04

a $x^2=3x$ **b** $h^2-10h-11=0$ **c** $5a^2=15a+20$

d $5x^2+7x-1=0$ **e** $y^2-6y=15$ **f** $8-5t-10t^2=0$

6 Solve each quadratic equation. 10.05

a $m^4-6m^2+8=0$ **b** $2y^4-5y^2-12=0$ **c** $x^6-7x^3-8=0$

7 Write a quadratic equation for each problem, then solve the problem. 10.06

a The product of 2 consecutive even numbers is 1088. Find the numbers.

b A rectangular garden bed is 8 m longer than it is wide. If the area of the garden bed is 308 m^2, find the dimensions of the garden bed.

8 Graph $y=2x^2+5x-7$, showing its x- and y-intercepts. 10.07

9 Find the vertex of the parabola with equation: 10.08

a $y=(x+5)^2+1$ **b** $y=-(x-7)^2+4$

c $y=x^2+6x$ by completing the square

10 Graph each quadratic equation, showing the x-intercepts (correct to one decimal place where necessary), the y-intercept, the equation of the axis of symmetry and the coordinates of the vertex. 10.08

a $y=x^2-2x-24$ **b** $y=-8x-3x^2$ **c** $y=5x^2+3x-8$

11 Find the point of intersection of the line: 10.10

a $y=x+3$ and parabola $y=2x^2$ **b** $y=3-2x$ and parabola $y=x^2+2x-2$

c $y=4-x$ and circle $x^2+y^2=16$ **d** $y=x-6$ and hyperbola $y=-\frac{8}{x}$

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